

Rishabh Verma

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EDUCATION

New York University

Master of Science - Mechatronics and Robotics GPA: 3.8/4.0

Sep 2023 - May 2025

Brooklyn, NY

Visvesvaraya National Institute of Technology

Bachelor of Technology - Mechanical Engineering GPA: 8.14/10

Jul 2019 - May 2023

Nagpur, India

PUBLICATIONS

A. Saviolo, N. Picello, **R. Verma**, J. Mao, & G. Loianno. “**Reactive Collision Avoidance for Safe Agile Navigation**”, IEEE International Conference on Robotics and Automation (ICRA), 2025

R Verma, J Phillip, M Kotambkar, “**From Concept to Reality: An In-Depth Review of Autonomous Driving Strategies and Their Implementation**”, International Conference on Mechanical Design and Manufacturing 2023

EXPERIENCE

BotCrew

Robotics Engineer

Feb 2025 - Present

[\[Website\]](#)

- Investigated optimization strategies for real-time object classification using YOLOv8 and TensorRT in C++, achieving significant inference speedups for embedded robotic systems.
- Designed and experimentally validated a Nonlinear Model Predictive Control (NMPC) framework for the flagship Gravion AGV Robot, enabling robust real-time trajectory tracking and adaptive motion planning.
- Proposed a scalable, ROS-agnostic mapping framework capable of handling large-scale (5,000 acres) spatial data, demonstrating improvements in computational efficiency and system scalability over existing frameworks such as NAV2.
- Implemented ORB-SLAM3 for RGB-D SLAM on the Gravion Robot, enabling accurate real-time localization and mapping for robust autonomous navigation in dynamic environments.

Agile Robotics and Perception Laboratory

Graduate Researcher

Sep 2023 - Jan 2025

[\[Website\]](#)

- Developed a novel reactive collision avoidance framework integrating perception, planning, and control, improving real-time obstacle detection and response time for quadrotors at speeds up to 7.2 m/s in diverse environments.
- Collaborated with the design team to optimize RACE platform, increasing flight time by 26% through firmware improvements, resulting in greater crash resilience and extended mission duration.
- Authored and presented research combining nonlinear model predictive control and adaptive control barrier functions, refining RGB-D data with neural networks to enhance depth accuracy and improve real-time obstacle avoidance.

Defence Research Development Organisation (DRDO)

Undergraduate Research Intern

May 2022 - Dec 2022

[\[Website\]](#)

- Developed a navigation pipeline designed to handle urban landscape complexities, which was rigorously tested across 15 different weather conditions and various maps, significantly enhancing overall system reliability.
- Implemented PID controller and A* navigation algorithms to enable precise control and navigation for the vehicle.
- Implemented YOLOv5 for pedestrian detection and achieved safe vehicle movement in 12 test cases of traffic situations avoiding collisions with pedestrians and vehicles.

SKILLS

Languages C, C++, CUDA, Python, Markdown

Frameworks ROS2, Gazebo, PX4, CARLA, PyTorch, TensorRT, CycloneDDS

Tools Git (Version Control), MATLAB, Simulink, CMake, Arduino IDE, NI Multisim, SolidWorks

KEY PROJECTS

Monocular Depth Estimation

[\[GitHub\]](#)

- Developed a lightweight monocular depth estimation model using a simplified encoder-decoder with skip connections, combining a truncated ResNet34 encoder and a bilinear-upsampled convolutional decoder.
- Achieved real-time inference at 30 FPS on an MIPI camera on an onboard Jetson Orin NX demonstrating suitability for real world applications.

State Estimation and Optical Flow-based Velocity Tracking

[\[GitHub\]](#)

- Developed a Vision based Pose and Velocity Estimation system for a quadcopter in indoor environment.
- Implemented Unscented Kalman Filter for State Estimation of a Quadcopter to capture non-linearity more accurately.

Robust Quadcopter Control in Windfield

[\[GitHub\]](#)

- Designed a control architecture for a quadcopter to reduce drift in variable windfield.
- Modeled the trajectory as an 8th degree spline for minimum snap constraints.
- Modeled the variable windfield as a form of Gaussian Noise and designed the controller to account for drag.
- Benchmarked the performance of PID, LQR, MPC, and Geometric Controller at cruising speeds upto 5 m/s.

COURSEWORK

Graduate

- Optimal Control and Reinforcement Learning
- Deep Learning
- Foundation of Robotics
- Mathematics for Robotics
- Robot Localization and Navigation

Undergraduate

- Machine Vision Applications
- Integral Transforms and Partial Differential Equations
- Single and Multivariable Calculus
- Matrix Theory
- Infinite Series and Ordinary Differential Equations